

Angela Chen

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Education

University of British Columbia

Sept 2024 – May 2029

Engineering Physics (GPA 4.0/4.33)

- **Relevant Coursework** Systems and Controls, Signals and Systems, Instrument Design, Software Construction, Experimental Techniques, Linear Circuits, Electricity and Magnetism, Mechanics, Applied Linear Algebra, ODEs, PDEs

Skills

Languages Python (NumPy, Flask, OpenCV), C/C++, Java, HTML, CSS, JS

Software & Tools Altium, KiCAD, Onshape, Git, Linux, MATLAB

Laboratory & Hardware Soldering, Oscilloscope, Multimeter, Function Generator, Logic Analyzer, 3D Printing, Raspberry Pi, ESP32, UART, I2C

Experience

UBC MINT (Multifaceted Innovation in Neurotechnology)

Jan 2025 – Present

Electrical Subteam Member

- Created a 4-channel EMG analog front-end circuit design for an EMG/EEG-controlled robotic arm, integrating INA333 instrumentation amps, two-stage $\sim 212\times$ gain, active Sallen-Key anti-aliasing filters, and a right-leg-drive circuit for active 60 Hz noise rejection into a 24-bit simultaneous-sampling SPI ADC.
 - Developing the biosignal-acquisition hardware end to end: schematic capture, single-supply 3.3 V / 1.65 V virtual-ground power architecture, component selection, and 4-layer PCB layout in Altium.
- Designed a 4-layer single-cell lithium-ion battery charger PCB in Altium while successfully addressing engineering considerations in thermal management, power integrity, grounding strategies and polygon placement.

Centre for Accessibility, University of British Columbia

Jan 2026 – May 2026

Designated Notetaker, Part-time Job

- Enhanced accessibility of a technical course (PHYS 301: Electricity & Magnetism) for a student with a disability by producing clear, comprehensive notes across the full academic semester.

Projects

Autonomous Robot, UBC ENPH 253 Mars Rover Competition

May 2026 – August 2026

- Co-designed and built a fully autonomous, 4-wheel omni-drive robot with a 3-processor (dual ESP32 + Raspberry Pi) control architecture, integrating closed-loop encoder/PID motor control, IR-based line following, and a multi-claw system designed specifically for each competition task, as part of a 4-person team.
- Co-owned overall electrical architecture and 4 custom 2-layer PCBs (with one partner) including comparator-based IR line-following sensor boards, discrete MOSFET H-bridge motor driver board, a multi-rail power distribution board, and drivetrain control logic (combinational H-bridge gate driving via 74HC-series gates, sensor MUXing).
- Developed firmware for the tape-following sensor array, including sensor driver code and line-position error calculation for closed-loop PID correction, and implemented portions of the robot's autonomous task sequencing logic.
- Built a real-time computer-vision system for an object detection task (Raspberry Pi 4 + ESP32): trained and validated a custom YOLOv26-nano object detector via transfer learning (~ 0.99 precision, 0.99 mAP@50), then optimized it for edge inference with an NCNN export, cutting Pi cold-boot time from ~ 67 s to ~ 7 s.
 - Engineered the surrounding embedded software: a multithreaded object detection state machine with multi-frame voting for robustness, a framed binary UART protocol (checksum + parser state machine) to interface with the dual ESP32, and reliable boot-on-power deployment as a Linux systemd service.

Custom FPV Drone, Personal Project

September 2026 – Present

- Designing a custom FPV quadcopter with STM32 flight controller and GPS in a 4-person group.

Awards

2026 UBC ENPH 253 Autonomous Robot Competition 3rd Place (out of 16 teams), Dean's List | **2025** Dean's List | **2024** District Authority Scholarship (Community Service), BC Achievement Scholarship, Lions Club Scholarship